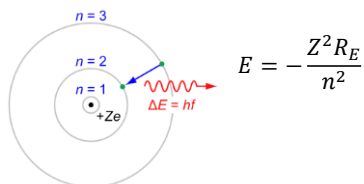


Optical properties of biological tissue on the microscopic level

Light absorption and emission

Recalling the Bohr model:

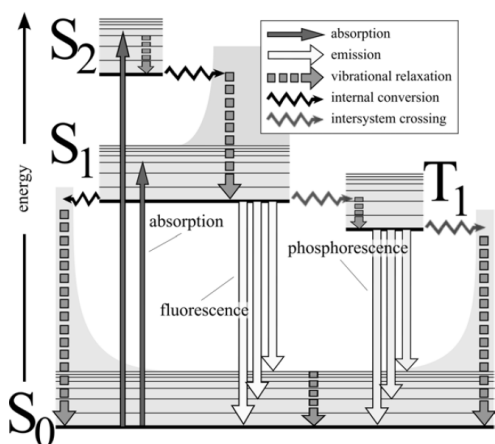


- ▶ Discrete energy level
- ▶ Transitions involve light absorption/emission

Additional effects in
the “real-world”:

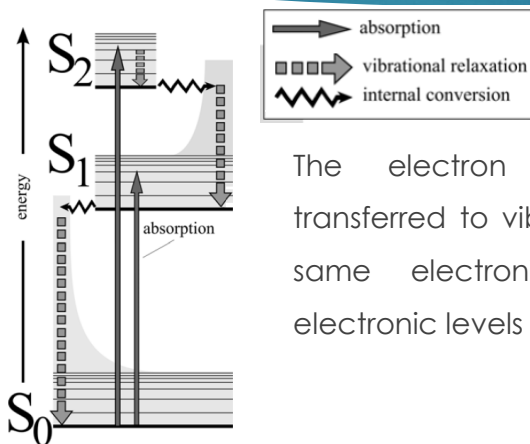
- ▶ Spin/angular momentum
- ▶ Vibrational modes
- ▶ Intermolecular interaction
- ▶ Temperature (phonons)

Jablonski diagrams



- ▶ Electronic transitions dominate energetically (more or less)
- ▶ Vibrational modes broadens the energetic levels
- ▶ Intersystem crossing enable forbidden transitions

Light turned into heat



The electron energy may be transferred to vibrational states in the same electronic level or lower electronic levels

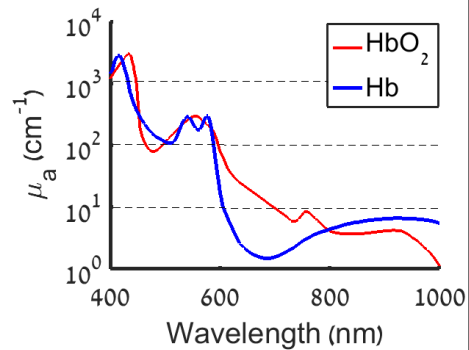
Time scales:

Absorption: $t \sim 1$ fs

Vibrational relaxation: $t \sim 1$ ps

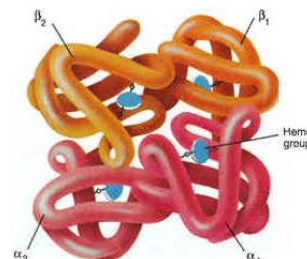
Blood

- ▶ Hematocrit of 45%: Blood is ~55% plasma (clear) and 45% hemoglobin (dark)
- ▶ Blood represents ~7% of body weight
- ▶ Binding to oxygen changes the absorption spectrum



Red blood cells and hemoglobin

- ▶ Red blood cells are in essence sacks of hemoglobin
- ▶ Hemoglobin has 4 sites to bind to oxygen (heme groups)



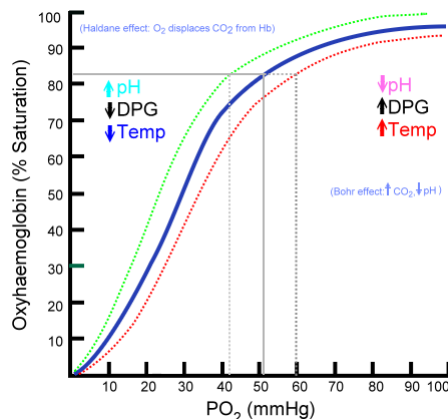
Oxygen saturation

Oxygen saturation (SO_2), i.e. $\frac{[HbO_2]}{[HbO_2] + [Hb]} \times 100$ can be calculated from the spectrum of the blood

Other useful definitions:

- ▶ SvO_2 : Venous oxygen saturation, 60%-80%
- ▶ SaO_2 : Arterial oxygen saturation, ~98%
- ▶ StO_2 : Tissue oxygen saturation

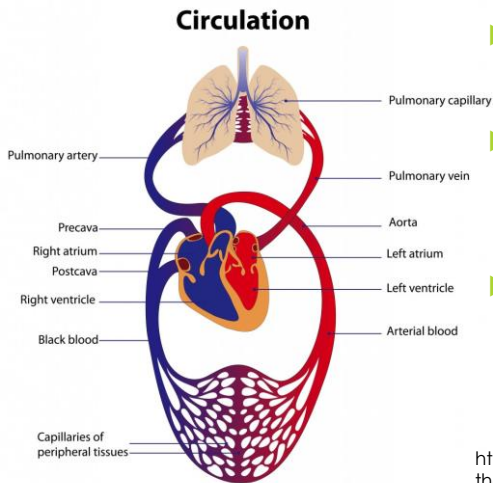
Oxygen-hemoglobin dissociation curve



PO_2 : partial pressure of oxygen in the blood – a measure for how much oxygen is dissolved in the blood plasma

https://en.wikipedia.org/wiki/Oxygen%E2%80%93hemoglobin_dissociation_curve

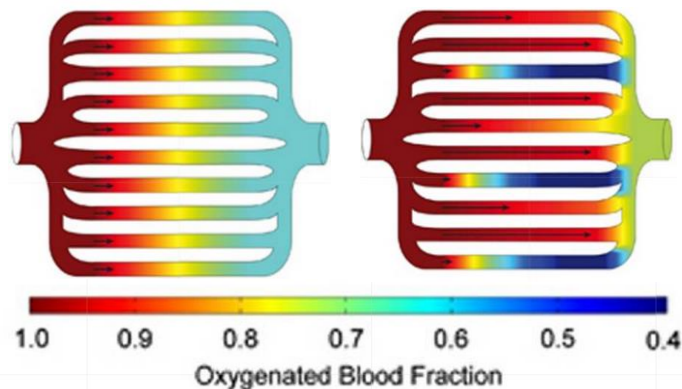
Blood circulation



- ▶ The capillaries are microscopic
- ▶ Macroscopically, in most organs, blood is "distributed" in the organ
- ▶ SO_2 of tissue is a combination of both arterial and venous blood

<http://medicalanatomy.net/labeled-diagram-of-the-circulatory-system/>

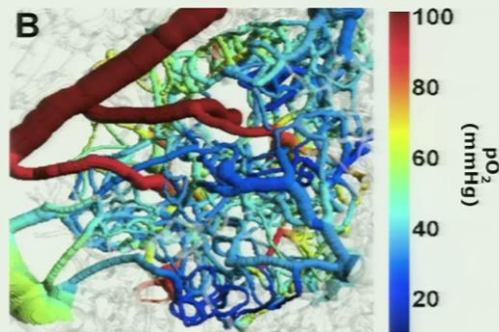
Circulation heterogeneity



Jespersen and Østergaard, J. Cereb Blood Flow Metab., **32**, 2012

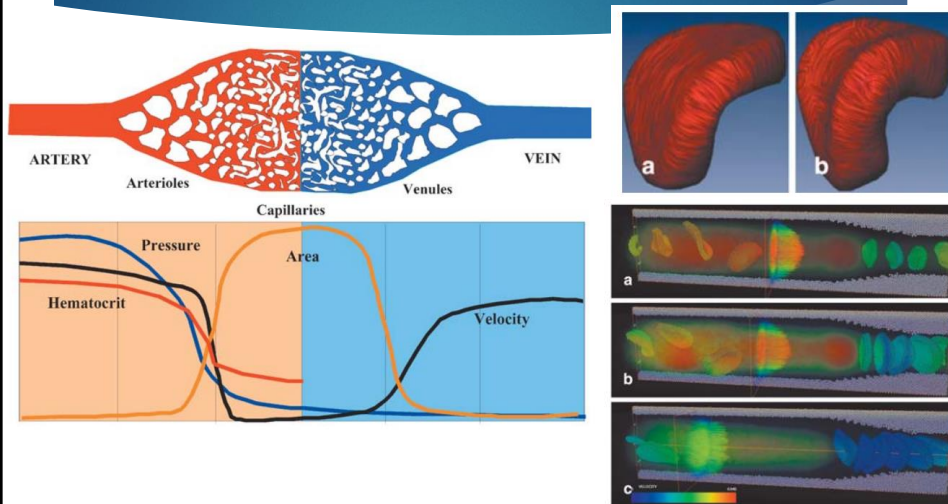
Circulation heterogeneity

Vascular Map of pO_2 Distribution



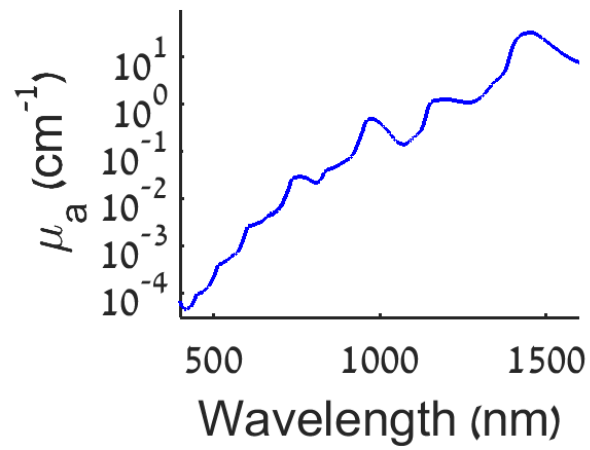
<https://www.youtube.com/watch?v=Q1juhgy-y9k>

Complex circulatory dynamics

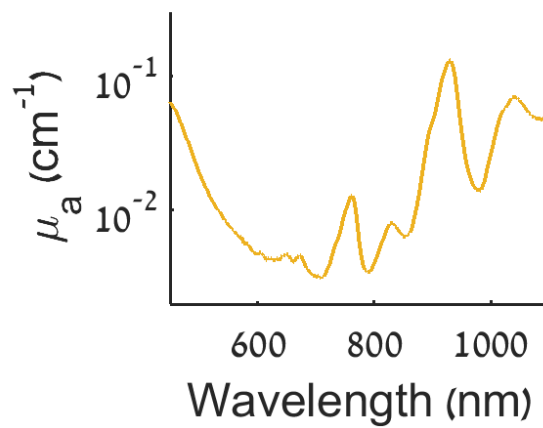


K. Boryczko et al. J. Mol. Model, **9** 16–33 (2003)

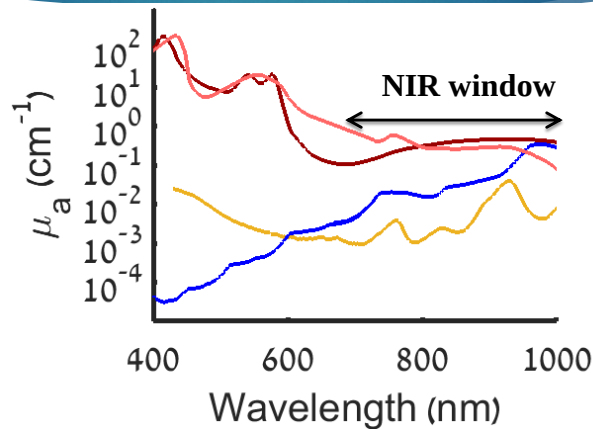
Water



Fat



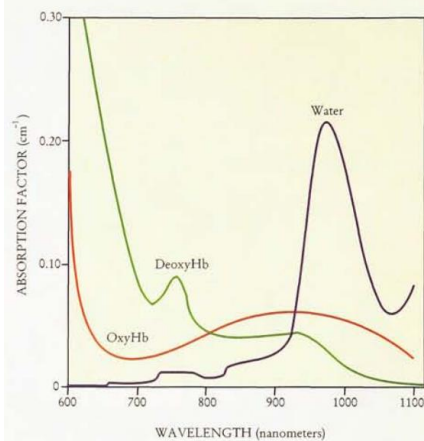
The near-infrared window



Assuming we are 65% water, 28% fat, and 7% blood (either entirely saturated or unsaturated)

The near-infrared window

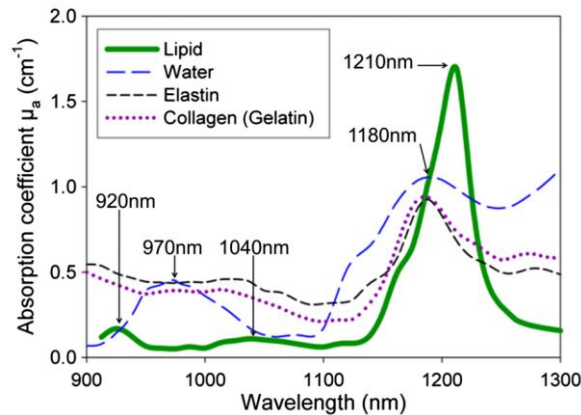
IN VITRO ABSORPTION SPECTRA of hemoglobin and water, showing a spectral window in tissue in the near-infrared. The window is brought about by a decrease in blood (oxy- and deoxyhemoglobin) absorption and an increase in water absorption. The concentrations of hemoglobin and water were adjusted to approximate values found in tissue such as the brain. These data were taken recently by Konda Reddy at the University of Pennsylvania. Absorptions are reported as natural logarithms. FIGURE 2



A. Yodh and B. Chance Phys. Today **48**, 1995.

Beyond the NIR window

Less penetration, but added ability in detecting additional tissue constituents



T. J. Allen et al. J. Bio. Opt. **17**, 2012.

Absorption heterogeneity

Topical Review

R57

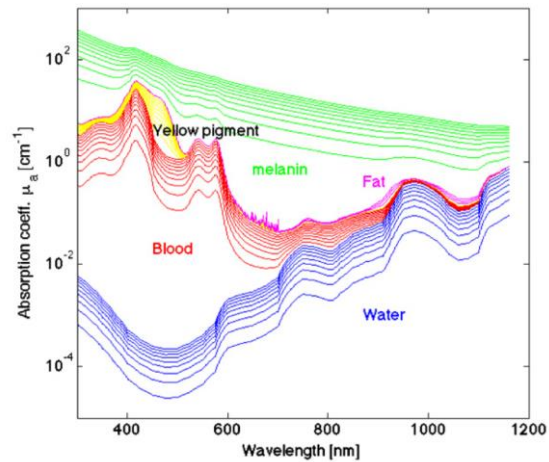
Table 3. *In vivo* tissue parameters governing optical absorption. C_{HGb} = total HGb concentration (μM), B = blood volume fraction $\times 100\%$ (assuming 150 g HGb L^{-1} blood), S = oxygen saturation of HGb $\times 100\%$, W = water volume fraction $\times 100\%$, M = melanosome volume fraction $\times 100\%$. Human tissues unless otherwise labeled. (na = not available.)

Tissue (reference)	C_{HGb} (μM)	$B\%$	$S\%$	$W\%$	$F\%$	$M\%$
1 Breast, normal (Tromberg <i>et al</i> 1997)	23.6	1.02	67.6	14.4	65.6	0
2 Breast, normal (Bevilacqua <i>et al</i> 2000)	24.2	1.04	75.5	29.2	51.7	0
3 Breast, normal (Durduran <i>et al</i> 2002)	34.0	1.46	68.0	na	na	0
4 Breast, normal (Jakubowski <i>et al</i> 2004)	16.0	0.69	62.6	6.0	74.0	0
5 Breast, normal (Spinelli <i>et al</i> 2004)	15.7	0.67	66.4	14.5	58.0	0
6 Breast, tumor (Jakubowski <i>et al</i> 2004)	41.0	1.76	61.1	41.0	39.0	0
7 Abdomen (Jakubowski <i>et al</i> 2004)	12.5	0.54	76.0	11.0	69.0	0
8 Dermis (Choudhury <i>et al</i> 2010)	4.7	0.20	39.0	65.0	0	0
9 Epidermis (Choudhury <i>et al</i> 2010)	0	0	0	na	na	2.50

"The optical properties of tissues should be regarded as variable from tissue to tissue, person to person and even time to time"

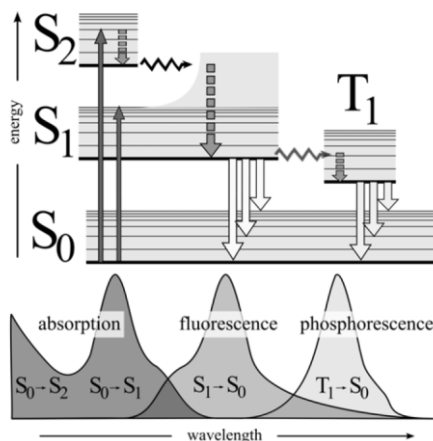
S. L. Jacques, Phys. Med. Biol. **58**, 2013.

Absorption heterogeneity



S. L. Jacques, Phys. Med. Biol. **58**, 2013.

Light turned into light



Time scales:

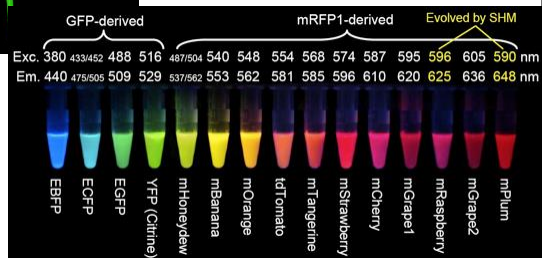
Absorption: $t \sim 1$ fs

Vibrational relaxation: $t \sim 1$ ps

Fluorescence: $t \sim 1$ ns

Phosphorescence: $t > 1$ μ s

Fluorescent proteins



http://www.nobelprize.org/nobel_prizes/chemistry/laureates/lmth.erutcel-neist/2008

Fluorescent proteins

Optical characteristics of the red-shifted FPs compared to GFP

	peak excitation	mol. ext. coeff	peak emission	quant. yield	relative brightness	excitation
	λ_{ex}^{max} (nm)	ϵ ($M^{-1}cm^{-1}$)	λ_{em} (nm)	q	B_r	λ_{ex} (nm)
GFP	489	53000	509	0.60	1	488
tdTomato	554	138000	581	0.69	2.99	488/532
mCherry	587	72000	610	0.22	0.49	532/594
mRaspb.	598	86000	625	0.15	0.40	594
mRFP	584	50000	607	0.25	0.39	594
mPlum	590	41000	649	0.10	0.12	594
Katashka	588	65000	635	0.34	0.67	594

N. C. Deliolanis et al. J Biomed Opt. **13**,2008

Exogenous fluorescent probes

Targeted

Non-targeted

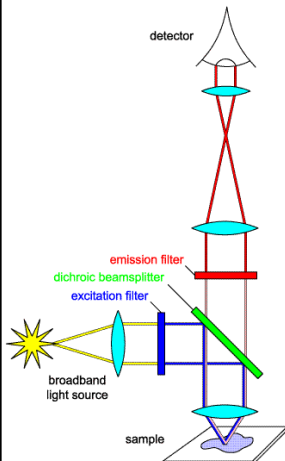
Activatable



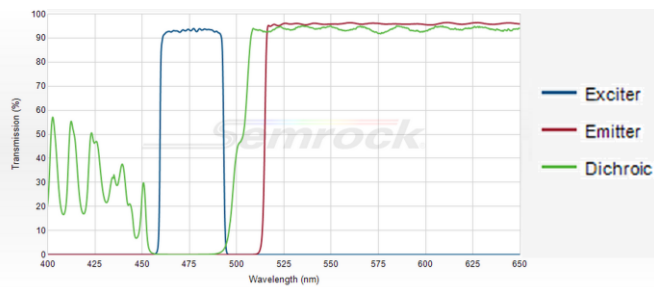
Auto-fluorescence

- ▶ Many biological structures have inherent capacity for fluorescence, e.g. elastin and collagen
- ▶ Usually auto-fluorescence is considered as non-specific background noise
- ▶ Nonetheless, analysis of auto-fluorescence may sometimes have diagnostic value

Fluorescence imaging: basic principles

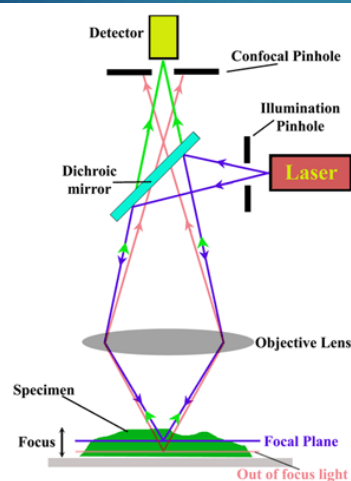


Filters are used to minimize cross-talk between excitation and emission



<http://www.semrock.com/>

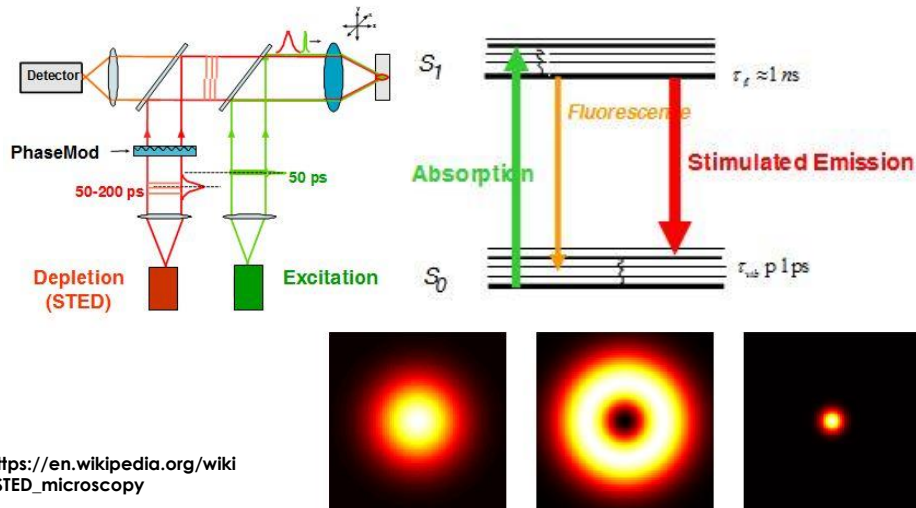
The confocal microscope



https://www.jic.ac.uk/microscopy/more/T5_8.htm

<http://www.leica-microsystems.com/science-lab/neuroscience-and-microscopy/>

Stimulated emission depletion (STED) microscopy



Bioluminescence

- Light emission via chemical reaction
- Same physical laws that determine fluorescence, but with no optical excitation
- No auto-fluorescence
- Complications in deep-tissue imaging (tomography)



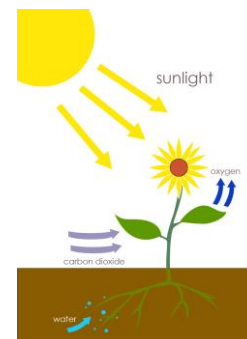
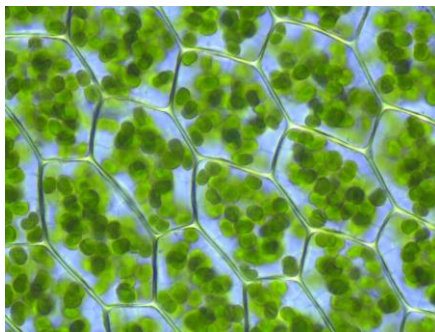
Thermal radiation

- ▶ Heat (either body heat or due to external illumination) leads to thermal radiation
- ▶ Body temperature corresponds to a wavelength of $9.5\ \mu\text{m}$ – not in the NIR window.



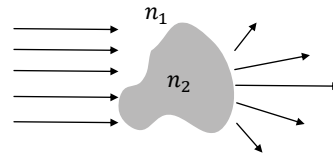
Light turned into chemical energy (food)

Like many other light-matter interaction, photosynthesis is not relevant to optical imaging in tissue



Scattering

- ▶ A result of variations in the refractive index in the medium
- ▶ We care about the statistical effect of many random scattered (no bandgap effects)
- ▶ Responsible for “whiteness” in our world: milk, cotton, white paint, white skin, white hair, etc.

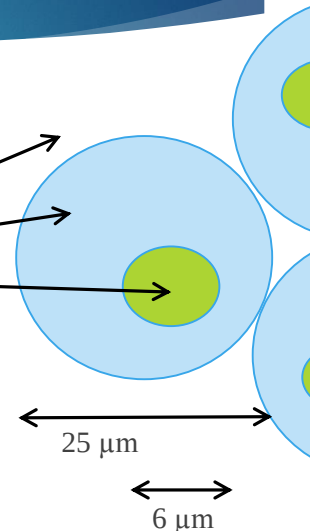


Scattering in tissue

Table 1.7 Characteristic Index of Refraction of Cellular Components (from [Drezek *et al.* (1999)])

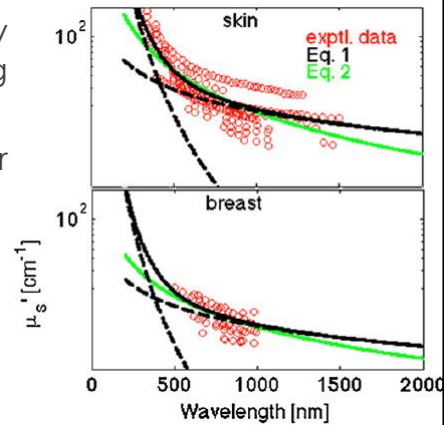
Structure	Index of Refraction
Extracellular Fluid	1.35-1.36
Cytoplasm	1.36-1.375
Nucleus	1.38-1.41
Mitochondria and Organelles	1.38-1.41
Melanin	1.6-1.7

- ▶ Each cell is almost transparent
- ▶ Main scatterers are larger than the wavelength



Scattering spectrum

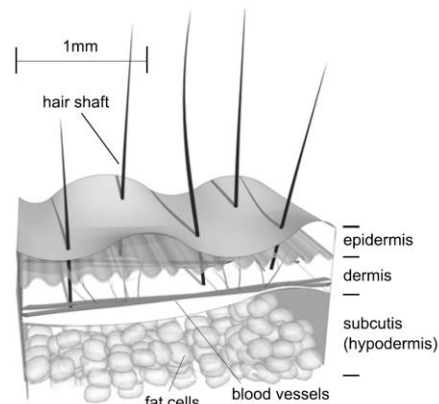
- Mathematically described by the reduced scattering coefficient μ'_s
- Spectral behavior determined empirically



S. L. Jacques, Phys. Med. Biol. **58**, 2013.

The effect of the epidermis

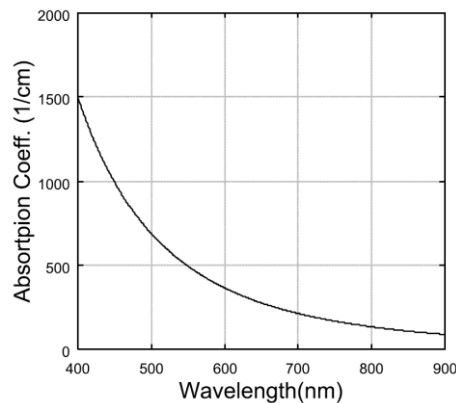
- Epidermis thickness ranges from 0.05 to 1.5 mm
- Avascular
- Main absorber is melanin



Absorption spectrum of melanin

In skin, the volume fraction of melanosomes is estimated to vary as

- ▶ Light skinned: $f_v = 1-3\%$
- ▶ Tanned: $f_v = 11-16\%$
- ▶ Dark skinned: $f_v = 18-43\%$



<http://omlc.org/spectra/melanin/>

Laser safety

Maximum permissible exposure (MPE): laser radiation to which a person may be exposed without hazardous effect or adverse biological changes in the eye or skin.

- ▶ A crucial parameter in designing imaging systems
- ▶ Depends on wavelength
- ▶ Depends on operation mode (CW, pulsed)

MPE: Example

Maximum Permissible Exposure (MPE) for Skin Exposure to a Laser Beam

Wavelength (μm)	Exposure Duration, t (s)	MPE	
		($\text{J} \cdot \text{cm}^{-2}$)	($\text{W} \cdot \text{cm}^{-2}$)
Visible and Near Infrared			
0.400 to 1.400	10^{-9} to 10^{-7}	$2 C_A \times 10^{-2}$	
	10^{-7} to 10	$1.1 C_A t^{0.25}$	
	10 to 3×10^4		$0.2 C_A$

Parameters/Correction Factors	Wavelength (μm)
$C_A = 1.0$	0.400 to 0.700
$C_A = 10^{2(\lambda - 0.700)}$	0.700 to 1.050
$C_A = 5.0$	1.050 to 1.400

ANSI Z136.1-2000